

Zero-Shot VLM Evaluation: Florence-2 Report

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1. Standard Performance Metrics

Total Evaluated Samples: 183 images

Success Rate ('Point' Prompt): 30.1%

Success Rate ('Grasp' Prompt): 30.6%

Per-Prompt Error ('Point' Prompt): 269.82 pixels

Per-Prompt Error ('Grasp' Prompt): 261.71 pixels

Prompt Sensitivity: 111.40 pixels

Prompt Consistency: 0.9200

Dual-Prompt Success Rate (Both <50px): 22.40%

2. Comparative Analysis: Per-Category Metrics

Legend: **Pt** = 'Point to...' Prompt | **Gr** = 'Grasp...' Prompt | **Px Err** = Pixel Error | **Norm Err** = Normalized Error | **Shift** = Prompt Sensitivity | **Succ** = Success Rate (<50px)

Category	Samples	Px Err (Pt)	Px Err (Gr)	Norm Err (Pt)	Norm Err (Gr)	Shift (px)	Succ (Pt)	Succ (Gr)
Toothbrush	4	280.5	490.8	22.8%	21.9%	582.0	25%	50%
Bag	4	1098.5	668.8	27.2%	17.9%	479.9	0%	0%
Highlighter	1	673.1	701.3	26.3%	27.4%	381.2	0%	0%
Tape	4	385.9	522.0	16.6%	21.7%	341.4	0%	0%
Spatula	4	268.6	182.8	28.2%	26.4%	235.2	25%	0%
Keys	4	92.2	313.0	10.3%	34.2%	234.7	50%	0%
Apple	4	539.0	476.8	29.1%	25.7%	218.5	0%	0%
Remote	4	131.5	321.5	11.4%	24.1%	209.7	75%	25%
Backpack	4	385.7	254.0	20.5%	12.7%	196.5	0%	50%
Calculator	4	198.0	163.4	12.0%	11.1%	193.0	0%	25%
Keyboard	4	113.4	299.4	8.6%	17.9%	186.9	75%	50%
Glove	4	173.3	115.2	18.3%	8.9%	151.9	25%	50%
Book	4	211.2	70.8	31.3%	11.2%	147.5	25%	50%
Watch	4	65.8	110.6	6.1%	10.6%	147.1	75%	50%
Fork	4	263.6	264.7	13.3%	13.7%	129.7	0%	0%
Mug	4	285.7	186.6	17.4%	11.5%	127.8	25%	25%
Wallet	4	313.5	243.6	16.7%	14.3%	120.5	25%	25%
Pot	4	485.1	467.5	29.6%	27.6%	119.6	0%	0%

Category	Samples	Px Err (Pt)	Px Err (Gr)	Norm Err (Pt)	Norm Err (Gr)	Shift (px)	Succ (Pt)	Succ (Gr)
Mouse	4	107.7	71.2	10.9%	5.6%	119.5	50%	75%
Shoe	4	46.6	66.8	7.7%	9.3%	109.4	75%	75%
Cup	4	825.1	764.8	21.6%	14.3%	86.2	50%	50%
Charger	4	574.5	605.0	25.0%	26.6%	84.8	25%	0%
Tray	4	510.0	476.3	20.9%	17.8%	81.9	25%	25%
Bowl	4	180.2	112.8	21.0%	13.5%	67.7	50%	50%
Scissor	4	349.4	322.6	29.3%	26.0%	67.1	0%	0%
Phone	4	94.5	40.2	5.7%	3.2%	65.1	50%	75%
Pen	4	74.5	119.6	4.4%	6.5%	58.8	50%	25%
Laptop	4	126.4	75.2	10.6%	4.7%	54.5	25%	25%
Knife	4	104.3	96.9	6.2%	5.1%	53.4	25%	50%
T-shirt	4	366.2	338.7	26.3%	24.2%	52.3	0%	25%
Spoon	4	121.5	74.0	12.8%	7.7%	47.7	25%	25%
Sock	4	60.7	105.4	8.9%	20.0%	47.0	75%	50%
Egg	4	89.0	49.7	9.6%	3.4%	45.4	25%	75%
Bottle	4	27.8	67.3	2.7%	8.8%	42.8	100%	75%
Umbrella	4	100.6	134.4	11.7%	15.9%	34.0	25%	25%
Basket	4	338.7	352.9	20.0%	20.8%	27.1	50%	50%
Kettle	4	253.5	253.8	23.9%	23.9%	13.3	50%	50%
Jar	4	435.5	426.3	32.9%	32.5%	10.0	0%	0%
Tomato	4	181.3	176.8	8.5%	8.3%	6.4	25%	25%
Stapler	3	166.5	165.8	17.9%	17.8%	1.7	33%	33%
Earbuds	2	266.7	266.6	22.4%	22.4%	0.8	0%	0%
Marker	4	70.5	70.5	12.2%	12.2%	0.7	25%	25%
Headphones	4	282.6	282.3	19.0%	19.0%	0.6	25%	25%
Pan	4	804.3	804.2	28.7%	28.7%	0.5	0%	0%
Notebook	4	107.0	107.1	6.8%	6.7%	0.5	50%	50%
Plate	4	186.3	186.1	16.0%	16.0%	0.4	25%	25%
Onion	4	145.7	145.7	23.2%	23.2%	0.3	0%	0%
Hairbrush	1	248.5	248.5	22.0%	22.0%	0.0	0%	0%
AGGREGATE	183	269.8	261.7	17.1%	16.4%	111.4	30.1%	30.6%

3. Qualitative Analysis: Raw Data Pipeline

Legend: *Size* = Image Dimensions (w,h) | *GT* = Ground Truth Coordinates | *Pt* = 'Point to...' Prediction | *Gr* = 'Grasp...' Prediction | *Px Err* = Pixel Error | *Shift* = Prompt Sensitivity

Image	Size (w,h)	GT (x,y)	Pt (x,y)	Gr (x,y)	Px Err (Pt)	Px Err (Gr)	Shift
bag_9.jpg	5312x2988	(788, 313)	(4236, 2079)	(2488, 1806)	3874.4	2263.0	1768.9
toothbrush_9.jpg	2048x1409	(1880, 182)	(1622, 188)	(58, 449)	258.4	1841.4	1585.3
tape_19.jpg	2448x2448	(866, 1436)	(1106, 1575)	(898, 761)	277.4	676.1	840.1
remote_13.jpg	1280x960	(966, 714)	(938, 716)	(229, 467)	27.4	776.8	751.6
keyboard_15.jpg	1600x1200	(28, 1094)	(45, 1101)	(781, 972)	18.0	762.5	747.1
spatula_12.jpg	1280x720	(652, 322)	(41, 587)	(628, 367)	665.8	51.5	626.8
apple_1.jpg	4608x3456	(2520, 777)	(3147, 1328)	(2656, 1626)	834.3	859.2	573.7
keys_5.jpg	705x940	(124, 670)	(124, 633)	(597, 321)	37.3	588.3	567.6
toothbrush_20.jpg	728x971	(210, 613)	(35, 97)	(210, 613)	544.4	0.4	544.5
watch_8.jpg	698x1241	(453, 923)	(411, 1112)	(360, 620)	194.2	316.7	494.7
wallet_3.jpg	2000x2000	(1086, 1425)	(520, 1017)	(1000, 1017)	698.2	417.4	480.0
backpack_20.jpg	1800x1119	(712, 496)	(579, 920)	(896, 580)	444.5	202.3	464.9
mug_4.jpg	1500x1125	(702, 855)	(1204, 504)	(789, 594)	612.2	275.0	425.0
calculator_11.jpg	828x1472	(641, 864)	(629, 977)	(415, 623)	113.6	330.8	413.9
highlighter_3.jpg	1536x2048	(1455, 1313)	(929, 893)	(758, 1234)	673.1	701.3	381.2
calculator_18.jpg	1728x1291	(809, 1007)	(412, 1005)	(769, 997)	396.8	41.3	356.9
fork_14.jpg	1320x1980	(1276, 1231)	(683, 715)	(968, 548)	786.8	750.0	330.6
tray_12.jpg	800x800	(306, 101)	(391, 392)	(469, 78)	302.8	164.8	323.8
tape_12.jpg	1280x720	(1173, 183)	(850, 417)	(541, 441)	399.3	683.0	310.0
apple_4.jpg	1600x1170	(690, 616)	(770, 229)	(776, 529)	395.1	121.7	300.2
cup_5.jpg	600x600	(249, 373)	(103, 84)	(259, 337)	324.5	37.8	297.6
mouse_15.jpg	640x1136	(291, 668)	(319, 878)	(550, 718)	212.4	263.5	280.9
pot_18.jpg	570x680	(423, 621)	(229, 201)	(159, 461)	462.4	309.1	269.6
bowl_3.jpg	750x500	(166, 331)	(682, 281)	(414, 305)	518.3	248.7	269.6
shoe_5.jpg	396x594	(95, 474)	(98, 473)	(356, 452)	3.9	261.8	258.0
glove_5.jpg	700x467	(473, 82)	(210, 436)	(256, 194)	441.4	244.3	247.1

Image	Size (w,h)	GT (x,y)	Pt (x,y)	Gr (x,y)	Px Err (Pt)	Px Err (Gr)	Shift
spatula_9.jpg	480x720	(109, 339)	(44, 384)	(91, 624)	78.5	285.7	245.1
backpack_14.jpg	1200x900	(806, 684)	(754, 461)	(796, 702)	229.5	19.4	244.4
pen_1.jpg	1920x1080	(881, 571)	(900, 602)	(666, 599)	36.6	217.0	234.3
book_6.jpg	337x500	(169, 169)	(235, 444)	(157, 229)	283.4	61.2	228.9
laptop_12.jpg	626x620	(430, 441)	(153, 492)	(369, 487)	281.4	76.7	216.0
tape_9.jpg	900x900	(350, 438)	(449, 214)	(458, 428)	244.6	107.8	213.9
phone_2.jpg	2048x1365	(892, 520)	(836, 742)	(867, 537)	229.1	30.9	207.0
charger_18.jpg	2220x1440	(768, 265)	(1300, 1030)	(1226, 1216)	931.9	1055.1	200.1
toothbrush_10.jpg	768x439	(238, 313)	(541, 232)	(351, 286)	313.2	116.0	197.3
mouse_8.jpg	634x476	(270, 197)	(103, 312)	(265, 202)	202.8	6.8	196.1
scissor_12.jpg	599x323	(407, 242)	(486, 75)	(306, 152)	184.6	135.5	195.9
keys_18.jpg	548x691	(236, 474)	(271, 474)	(270, 279)	34.4	197.4	194.9
book_18.jpg	400x400	(305, 60)	(50, 316)	(214, 215)	361.5	179.5	192.6
pot_5.jpg	768x576	(446, 525)	(352, 446)	(484, 310)	122.7	218.9	190.4